

This objective implements a trajectory-level policy gradient with group-centered, normalized advantages. The normalization by $\sum_j |A_j|$ rescales advantages within each group, reducing variance and preventing large score magnitudes from dominating the update. Formally, this follows the standard REINFORCE formulation with advantage baselines (Williams, 1992; Sutton and Barto, 2018), augmented with within-group normalization as in normalized and clipped advantage methods (Mnih, 2016; Schulman et al., 2017).

The use of group-relative advantages connects GRFR to rank-based and population-based optimization methods, including the cross-entropy method and evolution strategies (Rubinstein, 1999; Salimans, 2017), where learning is driven by relative performance within a cohort rather than by absolute returns. Unlike standard applications of these methods, GRFR applies group-relative policy gradients to rank *epistemic intervention strategies* rather than task-only trajectories, inducing a preference over full dialogue policies based on their long-horizon epistemic conduct.

Discussion. Group-Relative Frictive Ranking optimizes *relative epistemic quality* across full dialogue trajectories rather than absolute returns. By learning from comparative performance within each context, GRFR avoids dependence on the global calibration of the trajectory score and directly targets the ordering of epistemic strategies.

The main advantage of GRFR is that it captures long-horizon epistemic phenomena that cannot be reduced to local reward shaping or single-turn preferences: delayed clarification, cumulative contradiction, progressive misalignment, and escalation into unsafe regimes. Because updates are driven by relative ranking, GRFR remains robust to monotonic transformations of the trajectory score and to moderate miscalibration of the friction surrogate.

The primary cost of GRFR is computational: it requires grouped rollouts and trajectory-level evaluation, and its performance depends on sufficient diversity within each cohort. For these reasons, GRFR is best suited to agentic and multi-turn settings in which epistemic failures emerge only over extended interaction, and serves as the natural long-horizon extension of FAR and FPP.

6.5 Friction-Conditioned Trust Region (FTR)

The methods introduced so far incorporate epistemic friction into the *learning signal*, through

rewards, preferences, or rankings. Friction-Conditioned Trust Regions (FTR) instead operate at a different level of the learning pipeline: they regulate *how far the policy is allowed to move* from a trusted base model as a function of epistemic risk. Whereas the other FPO methods shape the learning objective, FTR shapes the feasible set of policies itself, making epistemic risk a first-class control variable over policy plasticity.

This idea builds on the central insight of trust-region and KL-regularized reinforcement learning: stable and safe policy optimization is achieved by constraining updates to remain close to a reference policy (Schulman et al., 2015, 2017; Achiam et al., 2017; Haarnoja et al., 2018). In standard formulations, the trust region radius is fixed globally. In FTR, we make this radius *context-dependent* and *risk-sensitive*, allowing the policy to be conservative in low-risk settings and plastic in high-risk ones.

Constraint. We impose a context-dependent trust region:

$$(26) \quad \text{KL}(\pi_\theta(\cdot | h) \parallel \pi_0(\cdot | h)) \leq \epsilon(h), \\ \text{where } \epsilon(h) = \epsilon_0 + \kappa \text{Risk}(h)$$

At first glance, this design appears to invert the standard logic of risk-sensitive control, where higher uncertainty typically calls for tighter trust regions and more conservative updates. The key distinction is that $\text{Risk}(h)$ in FTR does not represent uncertainty about the environment or the stability of learning dynamics, but epistemic danger of the *base policy itself* in context h .

When $\text{Risk}(h)$ is low, the base policy π_0 is judged reliable, and deviations are tightly constrained. When $\text{Risk}(h)$ is high, the base policy is precisely the source of potential harm: remaining close to π_0 would entrench unsafe, misleading, or normatively problematic behavior. In this regime, conservatism with respect to the prior becomes undesirable, and larger policy deviations are required to enable principled disobedience, refusal, or corrective intervention.

We note that an alternative design would shrink the trust region under high risk, preventing large updates in dangerous contexts. Such a choice is appropriate when risk reflects uncertainty about learning stability. In contrast, FTR is designed for settings where risk reflects the unreliability of the base policy itself, and where freezing the policy would perpetuate epistemic harm.

FTR thus implements *risk-conditioned plasticity*: it is conservative when the prior is trustworthy, and permissive when the prior is epistemically dangerous. This differs fundamentally from classical robust control, where trust regions shrink under uncertainty about dynamics; here, trust regions expand under lack of trust in the prior policy itself.

This constraint bounds the KL divergence between the updated policy π_θ and a base policy π_0 as a function of the estimated epistemic risk. When risk is low, the trust region radius $\epsilon(h)$ is small and updates are conservative; when risk is high, larger deviations are permitted, enabling risk-driven departures from the base behavior.

This form generalizes the fixed trust-region constraints used in TRPO and PPO (Schulman et al., 2015, 2017) by making the admissible update size an explicit function of epistemic danger. Conceptually, it implements *risk-conditioned policy plasticity*: the agent is encouraged to behave like its base model in safe, routine contexts, and is granted greater freedom to deviate only when epistemic risk warrants intervention.

Interpretation. When $\text{Risk}(h)$ is low, $\epsilon(h) \approx \epsilon_0$ and the update remains tightly anchored to π_0 . This prevents unnecessary changes to a well-aligned base policy in benign settings and preserves instruction-following and stylistic stability.

When $\text{Risk}(h)$ is high, $\epsilon(h)$ increases, relaxing the constraint and allowing the policy to deviate more aggressively from π_0 . This enables *principled disobedience*: the model may override its default behavior to clarify, refuse, defer, or otherwise intervene when the epistemic stakes are high.

Unlike reward-based or preference-based FPO methods, FTR does not require an explicit friction term in the objective. Instead, epistemic risk modulates the *geometry of policy updates* themselves, shaping where learning is allowed to occur rather than what is rewarded.

Optimization. This constraint can be enforced using standard techniques from constrained and KL-regularized policy optimization, including Lagrangian relaxation (Achiam et al., 2017) and PPO-style clipped or penalized objectives (Schulman et al., 2017; Haarnoja et al., 2018). In practice, $\text{Risk}(h)$ can be estimated by an auxiliary risk head and used to dynamically scale the KL penalty or clipping threshold during training.

Discussion. FTR is orthogonal to the other FPO methods. While FAR, FPP, and GRFR shape *what* the policy is optimized to prefer, FTR shapes *where* in policy space the optimizer is allowed to move. As a result, FTR can be combined with any of the other methods as a risk-sensitive regularizer.

Conceptually, FTR connects Frictive Policy Optimization to the literature on safe and constrained reinforcement learning (Achiam et al., 2017; Tamar et al., 2015; García and Fernández, 2015), while introducing a novel epistemic twist: safety constraints are not fixed, but adapt online to the model’s own uncertainty and epistemic danger estimates.

7 A Running Example Across FPO Methods

To make the friction functional and the FPO methods concrete, we use a single multimodal, multi-agent running example drawn from collaborative task-oriented dialogue (Khebour et al., 2024). The setting is a two-agent block-assembly task with shared visual state, similar to collaborative construction and weights tasks used in common ground tracking and multimodal dialogue research.

Let h_t denote the multimodal dialogue state at time t , consisting of:

- the linguistic dialogue history;
- a visual scene representation (block positions, colors, orientations);
- the task goal specification;
- each agent’s belief state over the partner’s intentions and knowledge.

At time t , the human partner issues the instruction:

“Place the red block on the tall stack next to the blue one.”

In the current scene, there are *two* tall stacks, and two blue blocks at different locations. The instruction is therefore referentially underspecified, and committing to an action without clarification risks task failure.

Candidate interventions. Consider three candidate intervention–utterance pairs (a, y) available to the agent:

- $(a_{\text{act}}, y_{\text{act}})$: directly execute a placement action on one hypothesized stack without clarification;

- $(a_{\text{clar}}, y_{\text{clar}})$: ask a clarifying question (e.g., “Do you mean the tall stack near the left blue block or the right one?”);
- $(a_{\text{repair}}, y_{\text{repair}})$: point out the ambiguity and propose a disambiguation using deictic reference (e.g., highlighting a candidate region in the visual scene).

These correspond to distinct epistemic intervention types in the taxonomy of Section 3.

Computing friction components. Using the decomposition in Section 5, we compute component frictions for each (h_t, y) :

- **Uncertainty/miscalibration.** $\text{Unc}(h_t, y_{\text{act}})$ is high because the agent commits despite high referential entropy; $\text{Unc}(h_t, y_{\text{clar}})$ is low because the agent defers commitment appropriately.
- **Contradiction.** $\text{Contr}(h_t, y)$ is low initially for all three, but may become high downstream if an incorrect placement contradicts later constraints.
- **Hazard.** Here hazard reflects task-level failure rather than physical danger: $\text{Haz}(h_t, y_{\text{act}})$ is high because an irreversible misplacement may corrupt the shared construction state.
- **Value conflict/intent forcing.** $\text{ValConf}(h_t, y_{\text{act}})$ is high because the agent silently fixes an interpretation of the partner’s intent; $\text{ValConf}(h_t, y_{\text{clar}})$ and $\text{ValConf}(h_t, y_{\text{repair}})$ are low.
- **Information gain.** $\text{InfoGain}(h_t, a_{\text{clar}})$ and $\text{InfoGain}(h_t, a_{\text{repair}})$ are high because they are expected to reduce referential uncertainty; $\text{InfoGain}(h_t, a_{\text{act}})$ is near zero.

Following the decomposition in Section 5, unproductive friction aggregates the four failure modes:

$$(27) \quad F^-(h, y) = w_1 \text{Unc}(h, y) + w_2 \text{Contr}(h, y) + w_3 \text{Haz}(h, y) + w_4 \text{ValConf}(h, y)$$

Productive friction captures information gain:

$$(28) \quad F^+(h_t, y) = w_5 \text{InfoGain}(h_t, a)$$

The net friction functional is therefore:

$$(29) \quad F(h_t, y) = F^+(h_t, y) - F^-(h_t, y)$$

Productive versus unproductive friction. Under the productive/unproductive decomposition:

- $F^+(h_t, y_{\text{clar}})$ and $F^+(h_t, y_{\text{repair}})$ are high because they reduce epistemic uncertainty;
- $F^-(h_t, y_{\text{act}})$ is high because it induces silent commitment and likely downstream repair cost.

Thus, in this context, clarification and repair constitute *productive friction*, while immediate action constitutes *unproductive friction*.

How the FPO methods treat the same example.

FAR (reward shaping). In Friction-Augmented Rewards, the high referential risk activates the gating function $g(\text{Risk}(h_t, y))$, so that

$$(30) \quad R'(h_t, a_{\text{clar}}, y_{\text{clar}}) > R'(h_t, a_{\text{act}}, y_{\text{act}})$$

even if immediate action appears to advance the task. The intervention cost $C_{\text{fric}}(a_{\text{clar}})$ prevents overuse of clarification in unambiguous scenes.

FPP (preference pairing). Friction Preference Pairing yields a preference pair:

$$(31) \quad \text{a. } y^+ = y_{\text{clar}} \quad \text{and} \quad y^- = y_{\text{act}}$$

whenever $F^+(h_t, y_{\text{clar}}) > F^+(h_t, y_{\text{act}})$ and $F^-(h_t, y_{\text{clar}}) < F^-(h_t, y_{\text{act}})$. Training therefore increases the probability of clarification in under-specified multimodal states.

GRFR (trajectory ranking). GRFR compares full dialogue–action trajectories:

- a trajectory that clarifies, resolves reference, and then places correctly;
- a trajectory that acts immediately, misplaces, and requires repair;
- a trajectory that repeatedly hesitates and stalls progress.

Ranking by $S(\tau)$ favors strategies that manage referential uncertainty early, minimizing downstream repair and task disruption.

FTR (risk-conditioned trust regions). Suppose the base policy π_0 is biased toward immediate action due to imitation of instruction-following data. In this ambiguous scene, $\text{Risk}(h_t)$ is high, so the admissible KL radius increases:

$$(32) \quad \text{KL}(\pi_\theta(\cdot | h_t) \| \pi_0(\cdot | h_t)) \leq \epsilon_0 + \kappa \text{Risk}(h_t)$$

This permits the optimizer to move probability mass away from a_{act} and toward a_{clar} or a_{repair} . In unambiguous scenes, the same mechanism keeps the policy tightly anchored to π_0 .